

1. JS for (let i = 2; i <= 50; i++) {
 console.log(i);
2. let num = 5; factorial = 1, temp = num;
 while (temp > 0) {
 factorial * = temp;
 temp--;
 }
 console.log('Factorial of {num} is: {factorial}');
3. let i = 20;
 do {
 console.log(i);
 i--;
 } while (i >= 1);
4. const str = "JavaScript";
 let reversedStr = "";
 for (let i = str.length - 1; i >= 0; i--) {
 reversedStr += str[i];
 }
 console.log(reversedStr);
5. const numbers = [45, 12, 89, 34, 67, 90, 23];
 let highest = numbers[0];
 for (let i = 1; i < numbers.length; i++) {
 if (numbers[i] > highest) {
 highest = numbers[i];
 }
 }
 console.log("Highest number: ", highest);

```

6. const numbers = [10, 15, 20, 25, 30, 35, 40];
let oddTotal = 0;
for (const num of numbers) {
  if (num % 2 !== 0) {
    console.log("odd number:", num);
    oddTotal += num;
  }
}
console.log("Total Sum of odd number:", oddTotal);

```

```

7. const student = {
  name: "Ravi",
  age: 22,
  course: "JavaScript",
  mark: 92
};
console.log('Name: ', { student: name }, Age: { student: age }, Course: {
  student: course }, Mark: { student: mark });

```

```

8. const students = [
  { name: "Arun", mark: 85 },
  { name: "Bala", mark: 85 },
  { name: "Chetan", mark: 90 },
  { name: "Deepak", mark: 70 }
];
for (const s of students) {
  if (s.mark > 75) {
    console.log(s.name, s.mark);
  }
}

```

```

9. // 1. named function
function SquareNormal (num) {
  return num * num;
}

```

11.2. Function Expression.

```
const squareExpression = function (num) {  
  return num * num;  
};
```

11.3. Arrow Function.

```
const squareArrow = num => num * num;
```

10. `const formatter = (name, age) => 'My name is {name} and I am {age} years old.'`

`console.log(formatter("Teja", 20));`

11. `{
 var a = "Var - function scope";
 let b = "let - Block scope";
 const c = "const - Block scope";
}`

`console.log(a); // console.log(b); // console.log(c);`

'1) Var - JS hoisted outside block)

'ii) b, c ReferenceError: b is not defined and c is not defined.'

12. $\rightarrow$ output with var: undefined

Reason: JS hoists variable declaration made with Var to the top of scope and init them undefined. The Value 10 is assigned only when the code reaches the declaration line.

assigned only when the code reaches the declaration line.

$\rightarrow$ output when replaced with let:

ReferenceError: cannot access 'a' before init

Reason: let declarations are hoisted into a "Temporal Dead Zone" (TDZ) and not init automatically. Preventing

access to the variable before its execution line.

```
13. const student = {  
  name: "Ravi",  
  age: 22,  
  course: "JS"  
};  
const { name, course } = student;  
console.log(name, course);
```

```
14. const SumAll = (...numbers) => numbers.reduce((total, num) =>  
  total + num, 0);  
console.log(SumAll(100, 20, 30, 40)); // output: 100
```

```
15. const users = [  
  { name: "Ravi",  
    mark: [80, 90, 85],  
    address: { city: "Chennai" }  
  },  
  {  
    name: "Kumar",  
    marks: [60, 70, 75]  
  }  
];
```

```
users.forEach(({ name, marks, address }) => {  
  const city = address?.city ?? "city not Available";  
  const totalMarks = marks.reduce((acc, cur) => acc + cur, 0);  
  console.log("Name: ", { name }, | city: { city } | Total marks:  
    { totalMarks });  
});
```